Abhinav Sharma

Applied AI/ML Engineer — LLM Agents · RAG · Backend

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SUMMARY

Applied AI engineer who ships LLM products end to end. Sole engineer on Zavyo, a multi-tenant WhatsApp SaaS with RAG chatbots and tool-calling AI agents, built and deployed in ~5 weeks (the company shut down before public launch). Background in deep learning and computer vision: behavioral cloning on a self-driving RC car and vision transformers on edge hardware.

EXPERIENCE

Synditech Technologies — SDE 1, Full-Stack & AI Engineer — Dec 2025 – May 2026

Zavyo: multi-tenant WhatsApp marketing & AI conversation SaaS (FastAPI, React 19, MongoDB, Qdrant, OpenAI) · *Showcase*

- Sole engineer: designed, built and deployed the platform on Meta's WhatsApp Cloud API, from first commit to a deployed, launch-ready build in ~5 weeks (400+ commits)
- Built tool-calling AI agents with per-bot configurable tools (sales handoff, external HTTP calls, product carousel, scheduled follow-ups), plus AI-based lead classification from conversations
- Built a RAG pipeline from scratch: PDF/DOCX/TXT/CSV ingestion, boundary-aware chunking, OpenAI embeddings, a separate Qdrant collection per tenant, and embedding usage metered against tenant credits
- Engineered race-safe, idempotent Razorpay billing: an atomic compare-and-set claim lock prevents double-crediting when webhooks and client verification arrive concurrently, across all payment paths
- Enforced tenant isolation across REST, WebSocket and vector layers; built multi-role RBAC with 50+ permission keys, per-plan resource caps, OTP throttling and login lockout
- Built a stateful router that executes no-code ReactFlow graphs as live WhatsApp conversations; 300+ API endpoints across 70+ collections, Dockerized and self-hosted via Coolify

OPEN SOURCE

LiteLLM (open-source LLM gateway) · PR #45358 — Oct 2026

- Fixed Vercel AI Gateway embedding models being registered as chat models, which broke proxy health checks; fixed the price-sync script and cost maps, added regression tests

PROJECTS

Spoken-English Grammar Scoring (SHL Hiring Assessment, Kaggle) — Whisper, DeBERTa-v3, Qwen2.5-7B, scikit-learn · GitHub

- Scores spoken-English grammar (1–5) from 45–60 s clips: normal + prompted-verbatim Whisper passes (their edit distance is a feature, r = −0.39), CoLA features, an LLM rubric grade and fine-tuned DeBERTa, blended on 5-fold CV (OOF RMSE 0.510; public leaderboard 0.394)

Offline RAG Stack: Air-Gapped AI Deployment — FastAPI, Ollama/vLLM, Qdrant, Docker · GitHub

- Self-hosted RAG system with zero internet egress: all services on an internal Docker network behind a firewalled edge relay (NET_ADMIN-only), with an automated air-gap check proving DNS, TCP and HTTP egress are blocked
- Offline install path (image and model bundle with SHA256 verification), Ollama/vLLM switchable by config, typed 504/502 failure mapping for model and vector-store outages; 77 tests, strict mypy, CI

Quotation Generation Agent — React, FastAPI, OpenAI · GitHub

- LLM agent that turns multi-PDF price sheets and customer requirement emails into structured quotations, flagging missing information
- Dual extraction (local PDF parser or LLM structured extraction, with automatic fallback), 3-tier JSON recovery (parse → retry → fallback), behavior governed by an editable agent.md spec, hard context guardrail, CI with GitHub Actions

Self-Driving RC Car — Behavioral Cloning — TensorFlow/Keras, OpenCV, Raspberry Pi, FastAPI · GitHub — Dec 2024 – Apr 2025

- Built the full pipeline on a custom Ackermann-steering car: collected ~4,500 labeled image–steering frames (3,700 train + 865 validation) with a multi-process data-collection rig (3 workers + shared memory)
- Trained a modified NVIDIA PilotNet CNN (8-class discretized steering) with on-the-fly augmentation; ran real-time inference on a laptop over a WebSocket link to the Pi's L298N motor control

SKILLS

AI/ML: LLM agents & tool-calling, RAG, OpenAI API, Ollama, PyTorch, TensorFlow/Keras, Hugging Face Transformers, CNNs, Vision Transformers, OpenCV

Backend: Python, FastAPI, Flask, REST, WebSockets, MongoDB, Qdrant

Infra: Docker, Coolify, GitHub Actions, AWS (Lambda, S3), Linux, Git

Frontend: React, JavaScript

EDUCATION & LEADERSHIP

Vellore Institute of Technology, Bhopal — B.Tech, Computer Science and Engineering — 2022 – 2026

Leadership: Event Management Lead, Open Source Club (Jan 2024 – Dec 2025) · Lab Setup Lead, Digital Logic Design Lab (Feb 2025)